SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA1442COURSE NAME: Compiler Design for High Level Programming

COURSE OUTCOME COVERED

CO3: *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 60Total Marks:60

Annexure A

MCQ

Code of Conduct:

I Arun B (192421346) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action. ***Signature of the student: Arun B***

Criteria Weightage	Level 4 – Exemplary (50–60 Marks)	Level 3 – Proficient (40–49 Marks)	Level 2 – Developing (30–39 Marks)	Level 1 – Beginnin g (0–29 Marks)
Number of Correct Answers	100% 50 to 60	correct answers 40 to 49 correct answers	30 to 39 correct answers	0 to 29 correct answers
Accuracy 100% Excellent (90– 100%)				

Good (75–89%) Average (50–74%) Poor (Below 50%)

Course Faculty

Annexure A

MCQ

1. A compiler designer wants to attach semantic meaning to grammar productions in order to generate intermediate representations efficiently. Which compiler concept is most suitable for this purpose?

Lexical buffering
Associating semantic rules with grammar productions
Register allocation
Peephole optimization

1. While compiling an arithmetic expression, a compiler internally creates a hierarchical representation showing operators and operands. Which structure is being generated?

Stack frame
Queue structure
Syntax Tree
Linked List

1. A compiler evaluates attributes from child nodes before computing the parent node value during semantic analysis. Which type of attribute evaluation is taking place?

Parent-driven evaluation
Synthesized attribute evaluation
Inherited attribute evaluation
Dynamic attribute evaluation

1. A language processor uses LR parsing to evaluate semantic information during reductions. Which attribute grammar is naturally suitable in this case?

Context-free attribute grammar
Recursive attribute grammar
S-attributed definition
Circular attribute grammar

1. During semantic rule evaluation, a parser requires both parent information and child information while traversing grammar symbols from left to right. Which definition supports this requirement?

S-attributed definition
Static semantic definition
L-attributed definition
Ambiguous definition

1. A compiler engineer wants child nodes to be processed before the parent node during semantic evaluation of expressions. Which traversal strategy should be selected?

Preorder traversal
Inorder traversal
Postorder traversal
Level-order traversal

1. An expression such as `(a+b)*c` is represented internally without unnecessary grammar details. What does this representation mainly capture?

   Memory layout
   Hierarchical program structure
   Register allocation
   CPU execution order

1. While evaluating semantic rules, information is passed from a parent node toward its child nodes. Which type of attribute is involved?

   Synthesized attribute
   Terminal attribute
   Inherited attribute
   Dynamic attribute

1. A compiler rejects the statement `int x = "hello";` before execution begins. Which feature of the programming language is responsible for detecting this issue?

   Token generation
   Type system
   Syntax optimization
   Loop unrolling

1. Consider the statement `float value = 25;`. Which compiler activity occurs automatically here?

   Explicit casting

Dynamic conversion
Implicit type conversion
Symbol replacement

1. A semantic analyzer verifies whether variables are declared and operations are type-compatible. Which compiler phase performs these tasks?

Lexical Analysis
Syntax Analysis
Semantic Analysis
Linking

1. A compiler developer chooses a representation that eliminates unnecessary grammar symbols while preserving expression meaning. Which structure is preferred over a

parse tree?

Symbol Table
Syntax Tree
Token Stream
Finite Automaton

1. During compilation, all information regarding identifiers, scope, and datatype is maintained centrally. Which compiler component performs this responsibility?

Code Generator
Symbol Table
Syntax Parser
Optimizer

1. A compiler uses only synthesized information for semantic evaluation of grammar productions. Which type of attribute grammar is being implemented?

Dynamic grammar
L-attributed grammar
S-attributed grammar
Context-sensitive grammar

1. A recursive descent parser performs semantic translation while expanding productions from the start symbol downward. Which translation approach is being followed?

Bottom-up translation
Top-down translation
Shift-reduce translation
Operator precedence translation

1. During shift-reduce parsing, intermediate grammar symbols are temporarily stored before reductions occur. Which data structure supports this operation most effectively?

Queue
Heap
Stack
Linked List

1. A compiler reports an error when a boolean value is added to an integer variable. What compiler mechanism detected this problem?

Syntax checking
Type checking
Token validation
Memory management

1. A programmer declares a variable to store whole numbers without decimal values. Which datatype is most appropriate?

Boolean
Integer
Character
String

1. Two array types are considered equivalent because they possess identical internal structures even though their names differ. Which type equivalence principle is

applied?

Name equivalence
Structural equivalence
Functional equivalence
Dynamic equivalence

1. A compiler implementing LR parsing evaluates semantic rules during reductions. Which evaluation strategy is best suited for this implementation?

Top-down evaluation
Parallel evaluation
Bottom-up evaluation
Recursive evaluation

1. During expression analysis, a compiler stores operators as internal nodes and operands as leaves. Which representation is being used?

AVL Tree
Heap Tree
Syntax Tree
B-Tree

1. A semantic rule requires information from symbols appearing to the left side of a production. Which

attribute grammar supports such evaluation?

S-attributed grammar
L-attributed grammar
Ambiguous grammar
Recursive grammar

1. After semantic analysis, a compiler converts validated source constructs into machine-independent instructions. Which compiler phase performs this task?

Lexical Analysis
Intermediate Code Generation
Linking
Parsing

1. A programmer intentionally converts a floating-point value into an integer using a language construct. What is this process called?

Promotion
Reduction
Casting
Coercion

1. While scanning source code, a compiler groups characters into meaningful units such as keywords and identifiers. Which phase performs this operation?

Semantic Analysis
Syntax Analysis
Lexical Analysis
Optimization

1. A parse tree generated for a program statement explicitly contains all grammar symbols used in derivation. What is the primary characteristic of this structure?

Contains machine instructions
Represents grammar symbols
Removes terminals
Stores registers only

1. A bottom-up parser starts constructing derivations beginning with identifiers and constants before reaching the start symbol. Which nodes are processed first?

Root nodes
Internal nodes
Leaf nodes
Header nodes

1. During assignment checking, a compiler ensures the datatype of the expression matches the datatype of the variable receiving it. Which component performs this verification?

Loader
Semantic checker
Tokenizer
Syntax formatter

1. Two variables are treated as equivalent only because they were declared using the same type name. Which equivalence mechanism is being followed?

Structural equivalence

Dynamic equivalence
Name equivalence
Functional equivalence

1. A semantic evaluator computes all child expressions before processing the parent expression. Which traversal technique reflects this behavior?

Preorder traversal
Inorder traversal
Postorder traversal
Reverse traversal

1. A compiler designer inserts executable semantic instructions directly inside grammar productions to perform translation tasks. What is this mechanism called?

Syntax-directed translation scheme
Token generation mechanism
Recursive grammar scheme
DFA representation

1. A parser repeatedly performs shift and reduce operations until the start symbol is produced. Which parsing strategy is being used?

Predictive parsing
Top-down parsing
Bottom-up parsing
Recursive parsing

1. A strongly typed programming language prevents invalid arithmetic operations between incompatible datatypes. Which language feature enforces this?

Type system
Code optimizer

Symbol allocator
Loader

1. A programmer creates a collection capable of storing multiple values of the same datatype under one name. Which datatype category does this belong to?

Primitive datatype
Derived datatype
Boolean datatype
Character datatype

1. In a syntax tree representing an assignment statement, the highest node corresponds to the complete statement meaning. Which element does the root node generally

represent?

Constant value
Identifier name
Entire expression or statement
Memory address

1. A compiler reports an error because a variable is used outside its declared scope. Which compiler phase detects this issue?

Syntax Analysis
Semantic Analysis
Code Optimization
Linking

1. A compiler automatically converts an integer operand into a floating-point operand before arithmetic evaluation. What type of conversion is this?

Narrowing conversion

Implicit widening conversion
Explicit casting
Recursive conversion

1. During attribute evaluation, semantic rules are processed from left to right so that inherited information becomes available when required. Which attribute grammar follows this restriction?

Circular grammar
L-attributed grammar
S-attributed grammar
Ambiguous grammar

1. A compiler uses shift-reduce actions and reduction tables to parse complex programming constructs. Which parser is most likely being used?

Predictive parser
Recursive descent parser
LR parser
LL(1) parser

1. A compiler engineer associates semantic computations directly with grammar rules for translation purposes. To what are these semantic rules attached?

Tokens only
Grammar productions
Machine registers
Memory blocks

1. During semantic evaluation, a node computes its value entirely using values obtained from child nodes. Which attribute type is involved?

Inherited attribute
Dynamic attribute
Synthesized attribute

External attribute

1. A compiler designer prefers a compact internal representation that removes unnecessary non-terminal symbols from derivations. Which representation is best suited?

Parse Tree
Syntax Tree
NFA
Transition Graph

1. A program contains the statement `total = amount + tax;` where `tax` has never been declared. Which compiler phase identifies this error?

Lexical Analysis
Parsing
Semantic Analysis
Code Generation

1. A compiler automatically changes the datatype of an operand during expression evaluation to avoid type mismatch. What is this process called?

Coercion
Parsing
Reduction
Folding

1. Two structures are treated as equivalent because they contain identical field arrangements and datatypes. Which equivalence principle is being applied?

Name equivalence
Scope equivalence
Structural equivalence
Dynamic equivalence

1. A compiler implementing LR parsing evaluates semantic actions during reduce operations. Which evaluation strategy aligns best with this parser behavior?

Predictive evaluation
Recursive evaluation
Bottom-up evaluation
Top-down evaluation

1. A compiler developer chooses an attribute grammar that can be evaluated naturally during reduce actions without inherited information. Which grammar type is most appropriate?

L-attributed grammar
S-attributed grammar
Ambiguous grammar
Circular grammar

1. In a syntax tree for arithmetic expressions, the terminal values such as variables and constants appear at which location?

Internal nodes
Root node
Leaf nodes
Header nodes

1. A programmer explicitly converts a floating-point result into an integer before storing it in a variable. Which operation is being applied?

Promotion

Casting
Translation
Coercion

1. After successful syntax checking, a compiler begins verifying meanings, datatypes, and scope rules. Which phase starts next?

Lexical Analysis
Semantic Analysis
Linking
Loading

1. A programming language verifies datatype compatibility during compilation and sometimes again during execution. What does this indicate about type checking?

It occurs only during loading
It occurs only during runtime
It can occur during compile time and runtime
It is unnecessary in modern languages

1. A parser constructs derivations beginning from tokens and gradually reduces them until the start symbol is obtained. Which parsing approach is demonstrated?

Top-down parsing
Predictive parsing
Bottom-up parsing
Recursive parsing

1. A compiler internally represents expressions using parent-child relationships between operators and operands instead of linear text order. Which representation style is being followed?

Sequential representation
Hierarchical representation
Tabular representation
Circular representation

1. A compiler phase determines whether expressions, assignments, and declarations carry valid meaning according to language rules. Which aspect is primarily analyzed?

Syntax structure
Meaning of constructs
Register allocation
Instruction scheduling

1. Two datatype declarations are accepted as compatible because the language rules consider them logically interchangeable. What does this describe?

Operator precedence
Variable scope
Type equivalence
Memory allocation

Answers:

1. B) Associating semantic rules with grammar productions
2. C) Syntax Tree
3. B) Synthesized attribute evaluation
4. C) S-attributed definition
5. C) L-attributed definition
6. C) Postorder traversal
7. B) Hierarchical program structure
8. C) Inherited attribute
9. B) Type system
10. C) Implicit type conversion
11. C) Semantic Analysis

12. B) Syntax Tree
13. B) Symbol Table
14. C) S-attributed grammar
15. B) Top-down translation
16. C) Stack
17. B) Type checking
18. B) Integer
19. B) Structural equivalence
20. C) Bottom-up evaluation
21. C) Syntax Tree
22. B) L-attributed grammar
23. B) Intermediate Code Generation
24. C) Casting
25. C) Lexical Analysis
26. B) Represents grammar symbols
27. C) Leaf nodes
28. B) Semantic checker
29. C) Name equivalence
30. C) Postorder traversal
31. A) Syntax-directed translation scheme
32. C) Bottom-up parsing
33. A) Type system
34. B) Derived datatype
35. C) Entire expression or statement

1. B) Semantic Analysis
2. B) Implicit widening conversion
3. B) L-attributed grammar
4. C) LR parser
5. B) Grammar productions
6. C) Synthesized attribute
7. B) Syntax Tree
8. C) Semantic Analysis
9. A) Coercion
10. C) Structural equivalence
11. C) Bottom-up evaluation
12. B) S-attributed grammar
13. C) Leaf nodes
14. B) Casting
15. B) Semantic Analysis
16. C) It can occur during compile time and runtime
17. C) Bottom-up parsing
18. B) Hierarchical representation
19. B) Meaning of constructs
20. C) Type equivalence

RUBRICS FOR CALCULATION

Level 4 – Exemplary (90– 100 Marks) **Level 2 – Developing (50–74 Marks)**

Level 3 – Proficient (75–89 Marks) **Level 1 – Beginning (0– 49 Marks)**

Criteria Weightage

Number of 90 to 100 correct 75 to 89 correct 50 to 74 correct 0 to 49 Correct Answers 100% answers answers answers correct answers

Exemplary (5 Marks)

(4 Marks)

(2–3 Marks)

(0–1 Mark)

Basic

Demonstrate s clear and complete

Good understandi ng

understandi ng with some

Poor or incorrect

Conceptual Weight age (%)

Level 3 – Proficient

Level 2 – Developing

Level 1 – Beginning

Criteria

Level 4 –

Understandin g 30

partially correct understan ding

question

Mostly correct with minor errors

Covers all mistakes

Clear, concise, and parts with minor omissions

Incorrect or irrelevant answer

Accuracy of Answer 25

understandin g of the concept

Covers most application

No or incorrect application

Understand able with

Covers only some parts

Applies concepts with minor misconcepti ons

Completely correct answer with no errors

Incomplete or Response

15

Somewhat unclear or missing response

Application of Knowledge

Partially correct with noticeable errors

Correctly applies concepts to solve the question/prob lem

with minor gaps

Completenes s of

20

Clarity & Presentation 10

required parts of the

Difficult to understan d

Limited or

Criteria
Weight age (%)

Level 4 –
Exemplary (5

Marks)

well-organize d

answer

Level 3 –
Proficient

(4 Marks)

minor

issues
Level 2 –
Developing

(2–3 Marks)

poorly

structured
Level 1 –
Beginning

(0–1 Mark)